Business Finance Midterm Short-Answer Quiz: Concepts and Keywords

1. **[Ch.2]** What does a balance sheet report, how are its two sides ordered, and how is net working capital defined?

2. **[Ch.2]** Why are the book value and the market value of a firm's assets often very different, and which one matters more for decisions?

3. **[Ch.2]** Compare US GAAP and IFRS in terms of who sets them, their style, and how the order of the balance sheet differs.

4. **[Ch.2]** Define the average tax rate and the marginal tax rate. Which one is used for financial and investment decisions, and how do the two rates compare under a flat tax and under a progressive tax?

5. **[Ch.2]** Define cash flow from assets (CFFA). What are its three components, and what other two cash flows must it equal in total?

6. **[Ch.3]** What are common-size balance sheets and income statements, and why are they useful?

7. **[Ch.3]** List the five categories of financial ratios and give the question each category answers.

8. **[Ch.3]** State the DuPont identity and explain what each of its three components measures.

9. **[Ch.3]** Coca-Cola and PepsiCo both earn an ROE of about 40%. Using the DuPont figures in the table, explain how each firm reaches that ROE.

Item	Coca-Cola (KO)	PepsiCo (PEP)
Profit margin	27.34%	8.77%
Total asset turnover	0.46	0.87
Equity multiplier	3.26	5.26
ROE	40.7%	40.4%

10. **[Ch.3]** How can a ratio be benchmarked, and what problems can distort a comparison of ratios across firms?

11. **[Ch.4]** Why is a dollar today worth more than a dollar tomorrow? Distinguish simple from compound interest, and compounding from discounting.

12. **[Ch.4]** State the sign convention and the clearing rule for the TI BA II Plus time-value keys, and what happens if the signs are wrong.

13. **[Ch.5]** Define an annuity, an annuity due, a perpetuity and a growing perpetuity. What condition does the growing perpetuity need, and which cash flow goes in the numerator?

14. **[Ch.5]** Distinguish the APR from the EAR. Why must quoted rates be converted before comparing them, and what happens to the EAR as compounding becomes more frequent?

15. **[Ch.5]** In a loan repaid with equal payments, how is each payment divided between interest and principal over time, and how do you find the loan balance at any date?

16. **[Ch.5]** List common mistakes when solving annuity problems on the calculator or with the formula.

17. **[Ch.6]** Name the defining terms of a bond and say which are fixed at issue and which changes. How is a bond valued?

18. **[Ch.6]** How does the relation between coupon rate and yield to maturity determine whether a bond sells at par, a premium or a discount? How do maturity and coupon size affect interest rate risk?

19. **[Ch.6]** Define yield to maturity. What three assumptions lie behind it, and how do the coupon rate, current yield and YTM rank for a discount bond?

20. **[Ch.6]** What is a bond indenture, who enforces it, and what are the basic terms it contains? Why does a call provision lower a bond's value, and why do firms accept protective covenants?

21. **[Ch.6]** What do bond ratings measure, which is the lowest investment-grade rating, and what are lower-rated bonds called?

22. **[Ch.6]** State the Fisher effect and list the components that make up a corporate bond's nominal yield.

23. **[Ch.7]** Why is a stock harder to value than a bond? What is the value of a share, and how can a firm that pays no dividend still have value?

24. **[Ch.7]** Name the four dividend growth cases and state what to watch for in each.

25. **[Ch.7]** Split a stock's required return into its two components and describe each. Under constant growth, what is the capital gains yield?

26. **[Ch.7]** When are the P/E and price-sales ratios used to value a share, and why does the word "comparable" matter?

27. **[Ch.7]** Is preferred stock debt or equity? Describe its main features.

28. **[Ch.7]** Distinguish primary from secondary markets and a dealer from a broker. What is the bid-ask spread, and how does NASDAQ differ from the NYSE?

Answers

1. [Ch.2] The balance sheet is a **snapshot** of the firm's assets and liabilities at a given **point in time**, and it satisfies assets = liabilities + stockholders' **equity**. Assets are listed in order of **decreasing liquidity**; liabilities are listed in ascending order of **when due** to be paid. **Net working capital** = current assets − current liabilities, which is usually positive for a healthy firm.
Keywords: snapshot · decreasing liquidity · when due · net working capital · equity

2. [Ch.2] Book value is the accounting value shown on the balance sheet, based on **historical cost**. **Market value** is the price at which assets, liabilities or equity can actually be bought or sold, so it reflects the **future** cash-generating ability of the assets. The two differ because historical cost does not track current values or growth prospects, and the **market value** is what matters for financial **decision** making.
Keywords: historical cost · market value · future · decision

3. [Ch.2] US GAAP is set by the **FASB** (under SEC oversight) and is more **rules-based**. IFRS (including K-IFRS in Korea) is set by the **IASB** and is more **principles-based**. Both report the same core statements; the labels and layout differ, e.g. GAAP lists current (liquid) items first while IFRS lists **non-current** items first, so the order flips.
Keywords: FASB · IASB · rules-based · principles-based · non-current

4. [Ch.2] The average tax rate = **total taxes** ÷ taxable income, the overall tax bite. The marginal tax rate is the rate paid on the **next dollar** of income, and it is the one used for **decision** making. Under a **flat** tax the marginal rate equals the average rate (U.S. 21%); under a **progressive** schedule (e.g. Korea) the marginal rate is higher than the average rate.
Keywords: total taxes · next dollar · decision · flat · progressive

5. [Ch.2] CFFA = **operating cash flow** (OCF) − **net capital spending** (NCS) − the change in **net working capital**. It shows how cash is generated by using assets and how it is paid to those who finance them, so it equals cash flow to **creditors** plus cash flow to **stockholders**. This differs from the accounting statement of cash flows.
Keywords: operating cash flow · net capital spending · net working capital · creditors · stockholders

6. [Ch.3] On a common-size balance sheet every account is a **percent of total assets**; on a common-size income statement every line item is a **percent of sales**. Standardized statements make it easy to compare **year-to-year** results and to compare companies of **different sizes**, particularly within the same industry.
Keywords: percent of total assets · percent of sales · year-to-year · different sizes

7. [Ch.3] Liquidity (short-term solvency) ratios ask whether the firm can pay its short-term bills; financial **leverage** (long-term solvency) ratios ask how much debt is used; asset management or **turnover** ratios ask how efficiently assets are used; **profitability** ratios ask how much profit is earned; and **market value** ratios (P/E, M/B, EBITDA ratio) compare the stock price with accounting numbers.
Keywords: liquidity · leverage · turnover · profitability · market value

8. [Ch.3] ROE = **profit margin** × total **asset turnover** × **equity multiplier**. Profit margin (NI/Sales) measures profitability and **cost control**; total asset turnover (Sales/TA) measures asset-use **efficiency**; the equity multiplier (TA/TE = 1 + D/E) measures financial **leverage**.
Keywords: profit margin · asset turnover · equity multiplier · leverage · efficiency · cost control

9. [Ch.3] Coca-Cola gets there through a very high **profit margin** (27.3%, an asset-light concentrate model) but low asset turnover (0.46). **PepsiCo** has a thin margin (8.8%) and relies on faster **asset turnover** (0.87) and greater financial **leverage** (equity multiplier 5.26 vs 3.26). Same destination, different routes.
Keywords: Coca-Cola · PepsiCo · profit margin · asset turnover · leverage

10. [Ch.3] A ratio means little alone. Benchmark it with **time-trend** analysis (the firm's own history) or **peer group** analysis (similar firms or industry averages). Comparisons can mislead because of different **accounting methods** (FIFO/LIFO, depreciation), different **fiscal year** ends or seasonality, **conglomerates** with no clean peer group, differences in capital structure, and **window dressing** around reporting dates; there is no underlying theory giving a single correct value.
Keywords: time-trend · peer group · accounting methods · fiscal year · conglomerate · window dressing

11. [Ch.4] A dollar today can be **invested** to **earn a return**. **Simple interest** is paid only on the original principal, while compound interest pays **interest on interest**, so value grows faster and faster. Compounding multiplies by $(1 + r)^t$ to get a **future value**; discounting divides by $(1 + r)^t$ to get a **present value**.
Keywords: earn a return · simple interest · interest on interest · present value · future value

12. [Ch.4] Clear the TVM registers before every problem (2nd CLR TVM), because old values stay in memory. Cash paid out is **negative** and cash received is positive, so PV and FV need **opposite** signs; otherwise **Error 5** appears when solving for I/Y or N. Enter four of N, I/Y, PV, **PMT**, FV and press **CPT** on the fifth; PMT = 0 for a single lump sum.
Keywords: clear · opposite · CPT · PMT · Error 5

13. [Ch.5] An annuity is a **level** cash flow for a fixed number of periods; an annuity due pays at the **beginning** of each period, so its value = ordinary annuity × (1 + r). A perpetuity is a cash flow that **never ends** (PV = C / r). A growing perpetuity (PV = C_1 / (r − g)) requires r **greater than g**, and C_1 is the cash flow **one period from today**, not the one just paid.
Keywords: level · beginning · never ends · greater than g · one period from today

14. [Ch.5] The APR is a **quoted** rate equal to the periodic rate × m; the **effective annual rate** (EAR = $(1 + APR/m)^m$ − 1) is the rate actually earned or paid over a year. Two loans with the same APR can cost different amounts, so convert both to EARs before you **compare** them. With **more frequent** compounding the EAR is **higher** for the same APR (15% APR: 15.00% annual, 16.08% monthly, 16.18% daily).
Keywords: quoted · effective annual rate · compare · more frequent · higher

15. [Ch.5] Each payment covers **interest and principal**. The payment is **constant**, but **interest falls** each period because the balance falls, so the **principal rises**. The loan balance at any date equals the **present value of the remaining payments**, discounted at the loan rate.
Keywords: interest and principal · interest falls · principal rises · constant · present value of the remaining payments

16. [Ch.5] Leaving the calculator in **BGN** mode corrupts later answers by a factor of (1 + r). The rate and N must be per payment period: use the **rate per period** (not the APR) and the **number of payments**, not the number of years. Always **clear** the TVM registers, and give PV and PMT the correct **sign**.
Keywords: BGN · rate per period · number of payments · clear · sign

17. [Ch.6] A bond's **face value** (repaid at **maturity**) and its **coupon** (coupon rate × face value) are fixed at issue. The **yield to maturity**, the market's required return, **changes** every day with the **market**. Bond value = the **present value** of the coupon annuity plus the present value of the face value, so everything that happens after issue happens to the price.
Keywords: face value · coupon · maturity · yield to maturity · present value · changes

18. [Ch.6] If the coupon rate equals the yield the bond sells at **par**; if the coupon rate exceeds the yield it sells at a **premium**; if it is below the yield it sells at a **discount**. Prices and yields move in the **opposite** direction. Interest rate risk is greater for a **longer maturity** and for a **lower coupon** (a zero coupon bond is the extreme).
Keywords: premium · discount · par · inverse · longer maturity · lower coupon

19. [Ch.6] YTM is the single discount rate that sets the present value of the promised cash flows equal to the **market price**: the bond's **internal rate of return**. It assumes the bond is **held to maturity**, there is **no default**, and coupons are **reinvested** at the YTM. For a discount bond, coupon rate < **current yield** < YTM.
Keywords: internal rate of return · market price · held to maturity · no default · reinvested · coupon rate < current yield

20. [Ch.6] The **indenture** is the written agreement between the issuer and its creditors (face value, coupon, security, seniority, repayment, call provisions, covenants); a **trustee**, usually a bank, enforces it. A **sinking fund** retires the issue gradually. A **call** provision gives the issuer the option to buy the bonds back, usually when rates have fallen, so investors demand a **higher yield** (a lower value). Protective **covenants** limit the borrower (negative and positive) and firms accept them because the reduction in yield is worth more than the lost flexibility.
Keywords: indenture · trustee · sinking fund · call · higher yield · covenant

21. [Ch.6] Ratings assess the likelihood that the promised payments are made (**default** risk) and the protection creditors have if they are not. **Investment grade** runs from AAA/Aaa down to **BBB** (Baa3 / BBB−, the lowest investment-grade rating); below that the bonds are **speculative** grade, or junk.
Keywords: default · investment grade · speculative · BBB

22. [Ch.6] The **Fisher** effect is 1 + R = (1 + r)(1 + h): nominal yields rise with expected inflation. The nominal yield = the **real rate** + **inflation premium** + **interest rate risk premium** (the term structure) plus bond-specific premiums: **default risk premium**, taxability premium and **liquidity premium**.
Keywords: Fisher · real rate · inflation premium · interest rate risk premium · default risk premium · liquidity premium

23. [Ch.7] A dividend is paid at the board's **discretion** (a bond coupon is contractual), a stock has **no maturity** date, and both the required return and the growth rate must be **estimated**. The price of a share is the **present value of all future dividends**; "I buy it to sell higher" is contained in the model. A firm that pays nothing now must **eventually** pay, or its shares are worthless.
Keywords: discretion · no maturity · estimate · present value of all future dividends · eventually

24. [Ch.7] **Zero growth** is a perpetuity (preferred stock; $D_1 = D_0$). **Constant growth** is the dividend growth model $P_0 = D_1 / (R - g)$: use the **next dividend** D_1 and require **R > g**. **Nonconstant** growth is irregular and then constant: find $P_t = D_{t+1} / (R - g)$ and discount it by t. **Two-stage** growth is g_1 for t years and then g_2.
Keywords: zero growth · constant growth · nonconstant · two-stage · next dividend · R > g

25. [Ch.7] R = **dividend yield** + **capital gains yield**. The dividend yield is **next year's dividend** divided by today's price (D_1 / P_0), the analogue of a bond's current yield. Under constant growth the price grows at exactly **g**, so the capital gains yield equals the **growth rate** g.
Keywords: dividend yield · capital gains yield · growth rate · next year's dividend

26. [Ch.7] When there is no dividend to discount, price = EPS × a **benchmark price-earnings** ratio, or sales per share × a **price-sales** ratio, taken from **comparable** firms (same industry, size and prospects) or the firm's own history. With **negative earnings** a P/E is meaningless but sales are always positive. The answer is only as good as the word comparable.
Keywords: price-earnings · price-sales · negative earnings · comparable · benchmark

27. [Ch.7] Preferred stock looks like debt (a **fixed dividend**, no voting rights, ranked **ahead of common** stock and often callable), but it is equity: skipping a dividend is **not default**, dividends are **not tax-deductible**, and holders rank **behind creditors**. Most issues are **cumulative**, so arrears must be paid before common dividends resume.
Keywords: fixed dividend · not default · cumulative · ahead of common · behind creditors · not tax-deductible

28. [Ch.7] In the **primary** market the issuer sells securities for the first time and raises money; in the **secondary** market investors trade existing shares and the firm receives nothing. A **dealer** trades for its own account and bears inventory risk; a **broker** only matches trades for a commission. The dealer's **bid** is the price it pays, the ask is the price it charges, and the **spread** is its compensation. NASDAQ is a dealer market (a computer network of competing market makers); the NYSE is a hybrid auction market with DMMs, now largely electronic.
Keywords: primary · secondary · dealer · broker · bid · spread